Lecture # 30 (Last)

Date: 4th Dec 2025

Day: Thursday

most restricted

* Level 4 : (RL) → FA, RE

DFA NFA λ-NFA

(stack) * Level 3 : (CFL) → NPDA, DPDA, ~~CFG~~
CFG

context sensitive

* Level 1 : (CSL) → LBA (Linear Bounded automata ... (as a transducer))

* Level 0 : (unrestricted) (infinite storage).
TM

→ Recursive languages (TM decideable)
accept or reject (h_a, h_r)

partially decidable top! → Recursively enumerable Language
(R.E) (rejection → infinite loop)
TM acceptable or recognizable.

TM → UTM (universal turing machine)

is TM recognize

$A_{TM} = \{ \langle M, w \rangle \mid M$ is a TM and $w \in \Sigma^* \}$

Accept/Reject ? Halt TM

$H(M, w) \rightarrow M(w)$ if (M = = Reject / infinite loop)
else return

Date: Day:

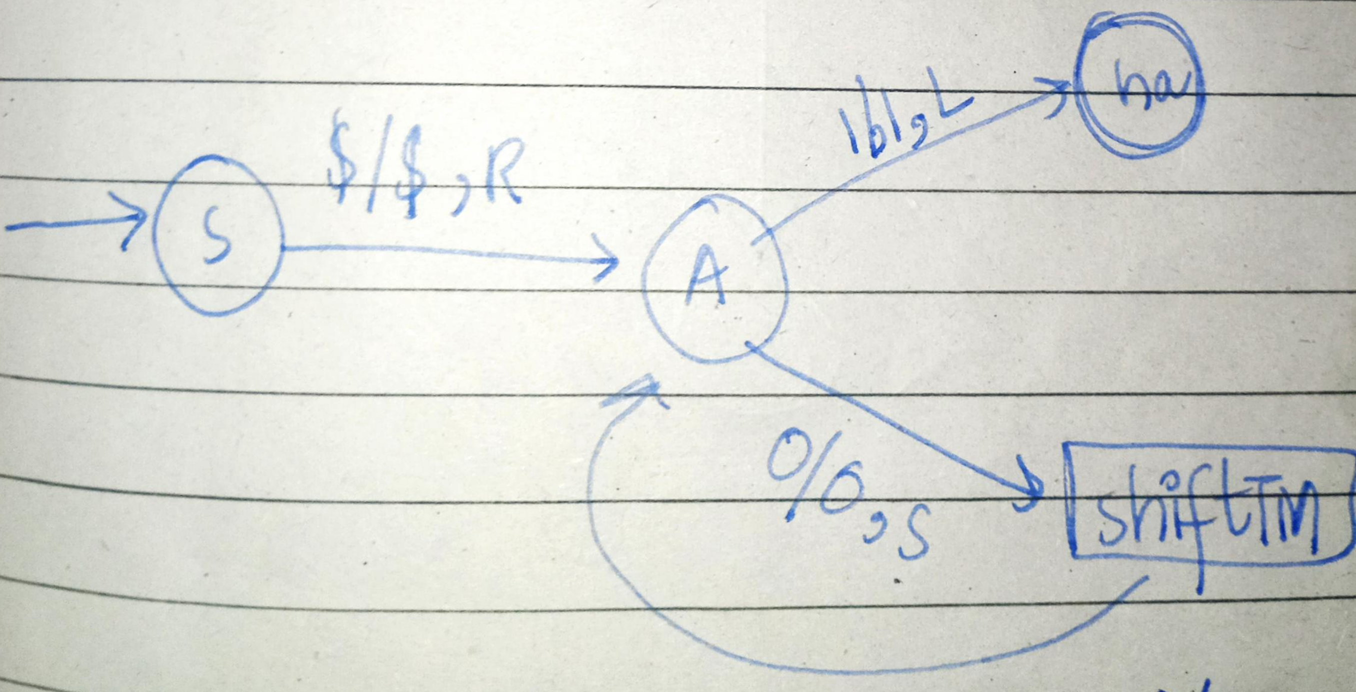
H

(M, w) (M, w)

halt not halt

not halt halt

\$	0	0	1	1	0	Δ



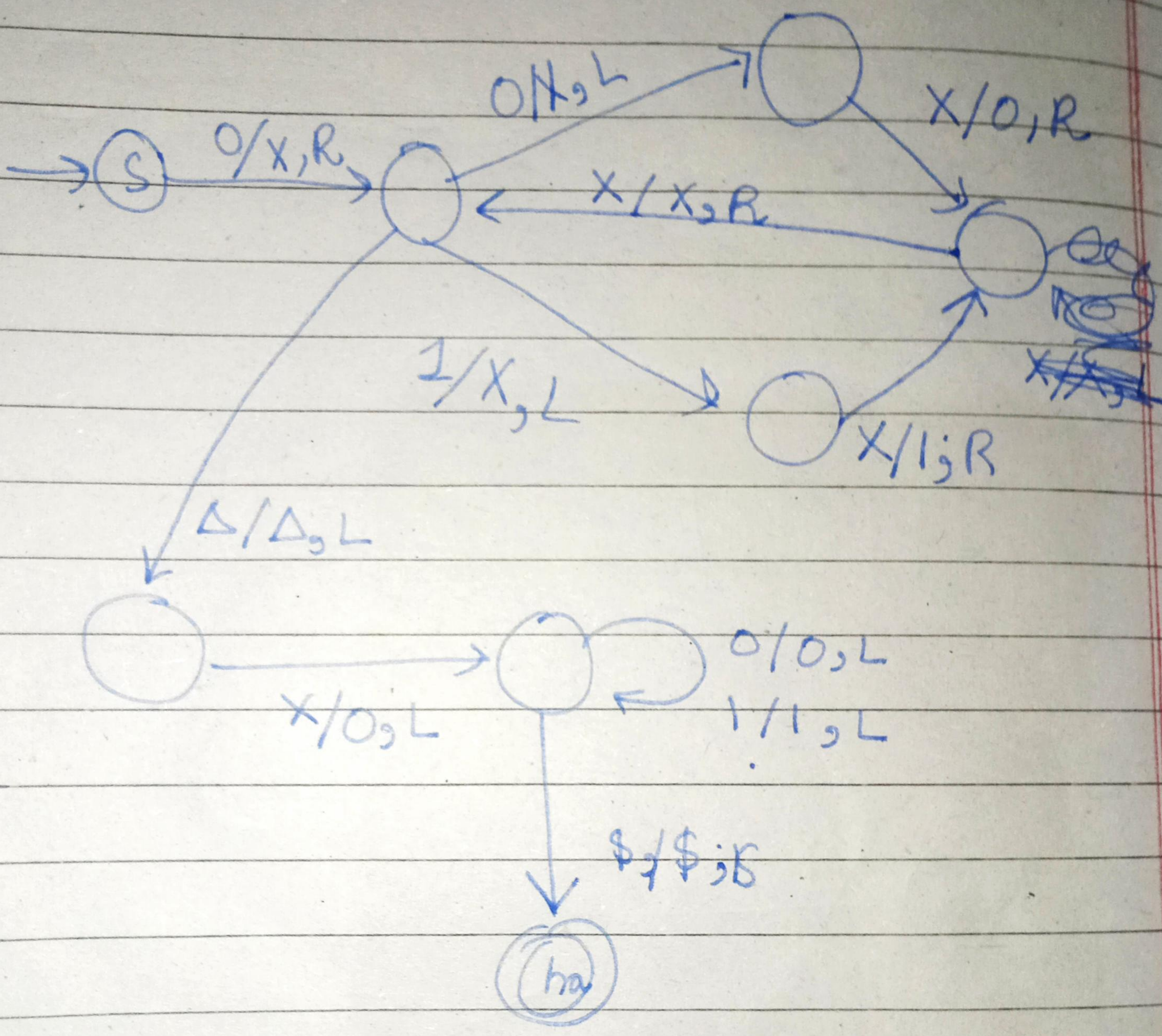
$	0	0	1	1	0	Δ

00110
00011
$0001

Date:

Day:

Left Shift



Good Luck Fellas!